

NORTHWIND SALES PERFORMANCE DASHBOARD

Using Microsoft Power BI

Project Report

Submitted By:

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Class: B.Com Fintech with AI

Tool Used: Microsoft Power BI Desktop

1. Introduction

Power BI is a business intelligence and data visualization tool developed by Microsoft. It helps users transform raw data into meaningful information through interactive reports and dashboards.

For this project, the **Northwind Sales Dataset** was used to create an interactive **Sales Performance Dashboard** using Microsoft Power BI. The dataset contains information about orders, customers, products, categories, employees and shippers.

The main purpose of the dashboard is to analyse sales and order performance, identify important product categories, understand customer distribution by country and compare the performance of employees. Interactive features such as slicers, KPI cards, charts, tables and tooltips are used to make the dashboard easy to explore.

2. Objectives of the Project

The major objectives of this project are:

- To analyse the overall sales and order performance of the business.
- To create an interactive dashboard using Microsoft Power BI.
- To analyse monthly order trends.
- To compare sales performance across different product categories.
- To identify the contribution of individual products to total sales.
- To analyse customers based on their countries.
- To compare the number of orders handled by employees.
- To analyse freight and shipping-related information.
- To provide interactive filtering using year, region and product category.
- To present business information in a simple and visual format.

3. Dataset Description

The project uses the **Northwind Sales Dataset**, which consists of multiple related tables. The major tables used in the Power BI report are:

Table	Description
Orders	Contains information about customer orders, order dates, employees, freight and shipping
Order Details	Contains product-level information such as quantity, unit price and discount
Products	Contains product names, prices and category information
Categories	Contains product category names and descriptions
Customers	Contains customer names, companies, cities and countries
Employees	Contains employee names and employee-related information
Shippers	Contains information about shipping companies

The dashboard uses these related tables to analyse sales, orders, products, customers and employees together.

Important information available in the dataset includes:

- Order ID
- Order Date
- Customer
- Employee
- Product
- Product Category
- Quantity
- Unit Price
- Discount
- Freight
- Country
- Region
- Shipper

4. Data Preparation

Before creating the dashboard, the data was prepared using **Power Query in Power BI**.

The main data preparation activities included:

- Importing the Northwind tables into Power BI Desktop.
- Checking the data types of all columns.
- Converting the Order Date field into the appropriate date format.
- Checking numerical fields such as Quantity, Unit Price and Freight.
- Checking categorical fields such as Product Name, Category Name, Country and Employee Name.
- Establishing relationships between the Orders, Order Details, Products, Categories, Customers, Employees and Shippers tables.
- Checking for missing or blank values.
- Preparing the data for creating visualizations.
- Creating DAX measures for the KPI cards.
- Creating interactive filters and tooltip pages.

Proper data preparation was important to ensure that the calculations and visualizations worked correctly.

5. DAX Measures Used

Several DAX measures were created for the dashboard.

→ Total Orders

This measure calculates the number of orders displayed in the dashboard.

Total Orders = `DISTINCTCOUNT(Orders[OrderID])`

Dashboard Result: 809

Purpose: It provides an overall view of the number of orders.

→ Total Quantity Sold

This measure calculates the total number of products sold.

Total Quantity Sold = `SUM('Order Details'[Quantity])`

Dashboard Result: 51K

Purpose: It shows the total sales volume in terms of units sold.

→ Total Freight

This measure calculates the total freight amount.

Total Freight = `SUM(Orders[Freight])`

Dashboard Result: 63.96K

Purpose: It provides an overview of the total freight associated with the orders.

→ Average Freight

This measure calculates the average freight per order.

Average Freight = `AVERAGE(Orders[Freight])`

Dashboard Result: 79.05

Purpose: It helps understand the average shipping/freight cost associated with orders.

→ Number of Customers

This measure calculates the number of customers included in the analysis.

Number of Customers = `DISTINCTCOUNT(Customers[CustomerID])`

Dashboard Result: 91

Purpose: It provides an overview of the customer base.

6. Dashboard Components

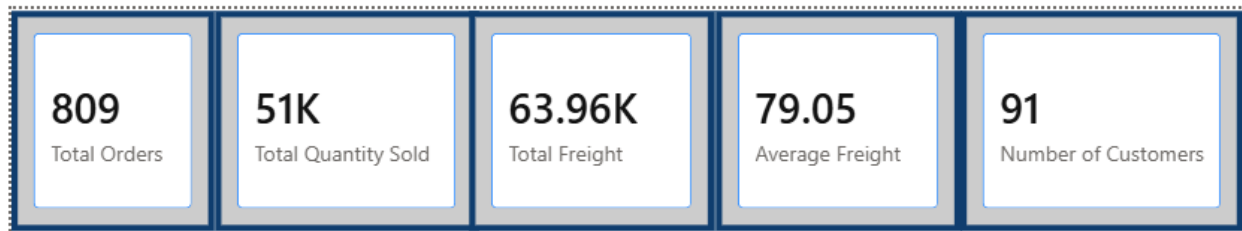
The Power BI project consists of multiple report pages, including **Sales Overview, Product & Customer Analysis, Interactive Sales Dashboard, Category Tooltip** and **Category Details**.

→ KPI Cards

Five KPI cards are displayed on the **Sales Overview** and **Interactive Sales Dashboard** pages:

- **Total Orders – 809**
- **Total Quantity Sold – 51K**
- **Total Freight – 63.96K**
- **Average Freight – 79.05**
- **Number of Customers – 91**

These KPI cards provide a quick summary of the overall business performance.



Purpose:

- To provide important information immediately.
- To summarize the main business indicators.
- To make the dashboard easier to understand.

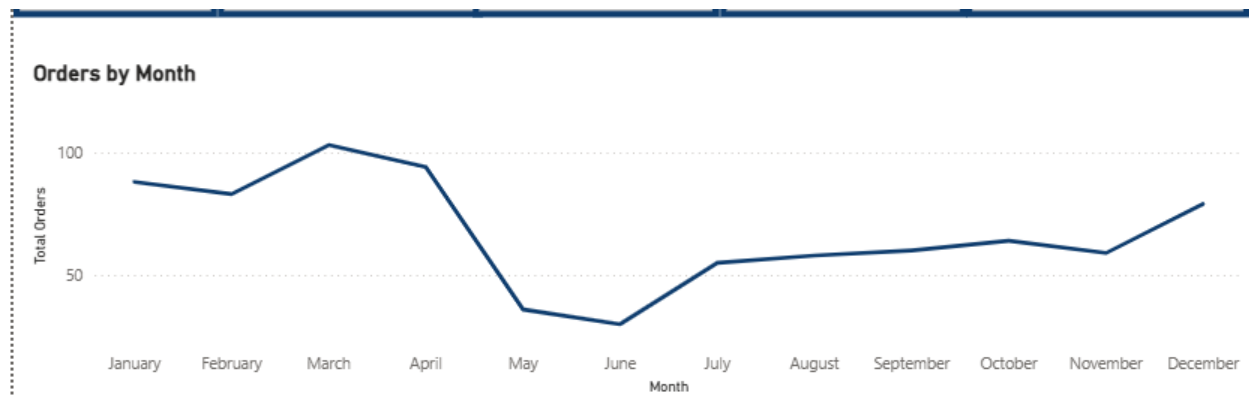
→ Orders by Month

A **Line Chart** is used to display the number of orders for each month.

The chart shows that the number of orders changes throughout the year. Orders are relatively high during the beginning of the year, reaching the highest level around **March**, followed by a significant decline during **May and June**. Orders then increase again from July onwards and reach a higher level in December.

Purpose:

- To identify monthly order trends.
- To identify high and low order periods.
- To understand seasonal changes in order activity.
- To support sales planning.



→ Orders by Employee

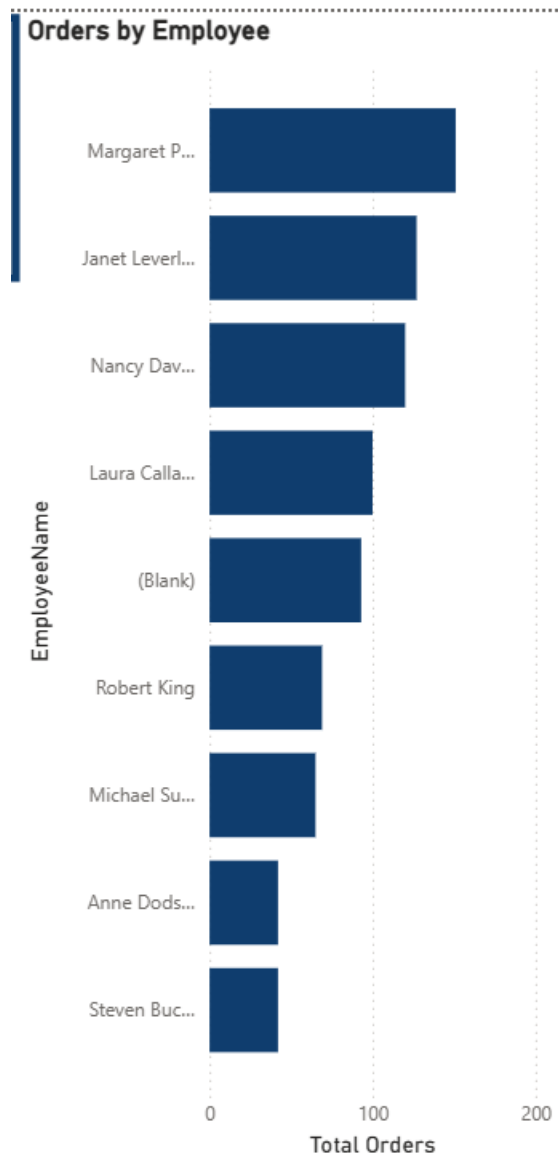
A **Horizontal Bar Chart** is used to compare the number of orders handled by different employees.

The chart shows differences in order volumes among employees. **Margaret Peacock** has the highest number of orders among the employees displayed, followed by **Janet Leverling** and **Nancy Davolio**.

A blank employee category is also displayed, representing records where an employee value is not available or does not match the displayed employee categories.

Purpose:

- To compare employee order performance.
- To identify employees handling higher order volumes.
- To understand the distribution of orders among employees.



→ Sales by Product Category

A **Column Chart** is used to compare total sales across product categories.

The dashboard includes the following categories:

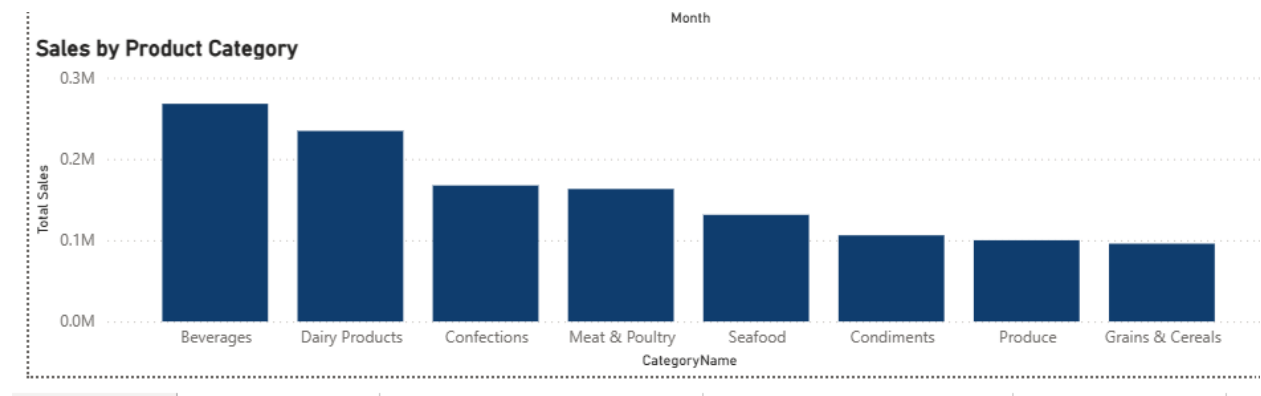
- Beverages
- Dairy Products
- Confections
- Meat & Poultry
- Seafood

- Condiments
- Produce
- Grains & Cereals

Beverages has the highest sales contribution, followed by **Dairy Products**. **Grains & Cereals** has the lowest sales among the categories shown.

Purpose:

- To compare category-wise sales.
- To identify high-performing product categories.
- To identify categories with comparatively lower sales.
- To understand product demand.



→ Product & Customer Analysis

A separate **Product & Customer Analysis** page provides detailed information about individual products.

A table is used to display:

- Product Name
- Category Name
- Total Sales
- Total Quantity Sold

For example, **Côte de Blaye** has total sales of approximately **141,396.74**, while the table also displays its quantity sold.

The table allows users to examine individual product performance in detail.

Purpose:

- To compare individual products.
- To identify high-sales products.
- To analyse product quantity sold.
- To provide detailed product-level information.

ProductName	CategoryName	Total Sales	Total Quantity Sold
Alice Mutton	Meat & Poultry	32,698.38	978
Aniseed Syrup	Condiments	3,044.00	328
Boston Crab Meat	Seafood	17,910.63	1103
Camembert Pierrot	Dairy Products	46,825.48	1577
Carnarvon Tigers	Seafood	29,171.88	539
Chai	Beverages	12,788.10	828
Chang	Beverages	16,355.96	1057
Chartreuse verte	Beverages	12,294.54	793
Chef Anton's Cajun Seasoning	Condiments	8,567.90	453
Chef Anton's Gumbo Mix	Condiments	5,347.20	298
Chocolade	Confections	1,368.71	138
Côte de Blaye	Beverages	1,41,396.74	623
Escargots de Bourgogne	Seafood	5,881.68	534
Filo Mix	Grains & Cereals	3,232.95	500
Flotemysost	Dairy Products	19,551.03	1057
Geitost	Dairy Products	1,648.13	755
Genen Shouyu	Condiments	1,784.83	122
Gnocchi di nonna Alice	Grains & Cereals	42,593.06	1263
Gorgonzola Telino	Dairy Products	14,920.88	1397
Grandma's Boysenberry Spread	Condiments	7,137.00	301
Gravad lax	Seafood	2,688.40	125
Guarana Fantastica	Beverages	4,504.37	1125
Gudbrandsdalsost	Dairy Products	21,942.36	714
Gula Malacca	Condiments	9,915.95	601
Gumbär Gummibärchen	Confections	19,849.14	753
Gustaf's Knackebröd	Grains & Cereals	7,122.36	348
Ikura	Seafood	20,867.34	742
Inlagd Sill	Seafood	13,458.46	805
Ipoh Coffee	Beverages	23,526.70	580
Jack's New England Clam Chowder	Seafood	8,680.35	981
Total		12,65,793.04	51317

→ Customers by Country

A **Horizontal Bar Chart** is used to display the number of customers by country.

The dashboard shows countries such as:

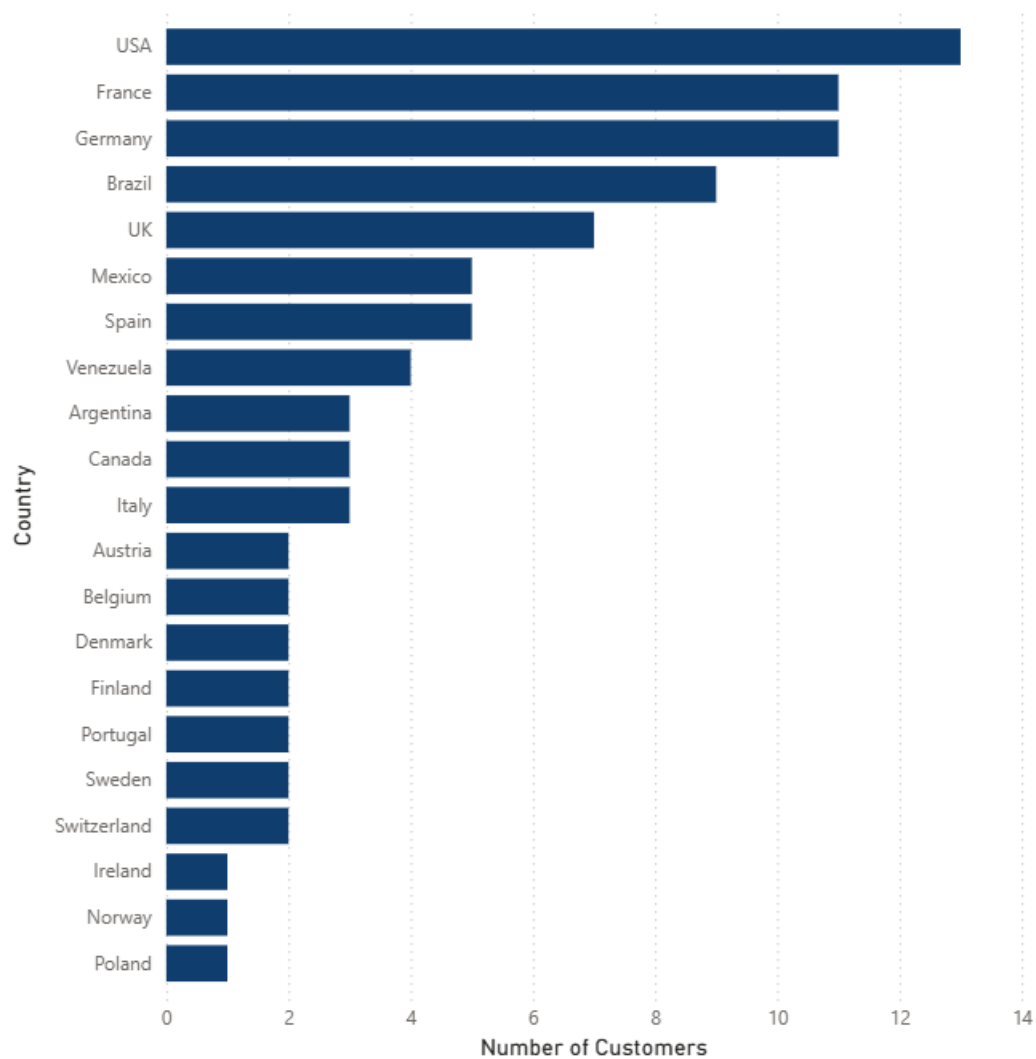
- USA
- France
- Germany
- Brazil
- UK
- Mexico
- Spain
- Venezuela
- Argentina
- Canada
- Italy
- Austria
- Belgium
- Denmark
- Finland
- Portugal
- Sweden
- Switzerland
- Ireland
- Norway
- Poland

The **USA** has the highest number of customers among the countries displayed.

Purpose:

- To analyse customer distribution by country.
- To identify major customer markets.
- To compare the size of customer bases across countries.
- To understand geographical customer distribution.

Customers by Country



→ Interactive Sales Dashboard

The **Interactive Sales Dashboard** combines multiple visualizations into a single interactive page.

The page contains:

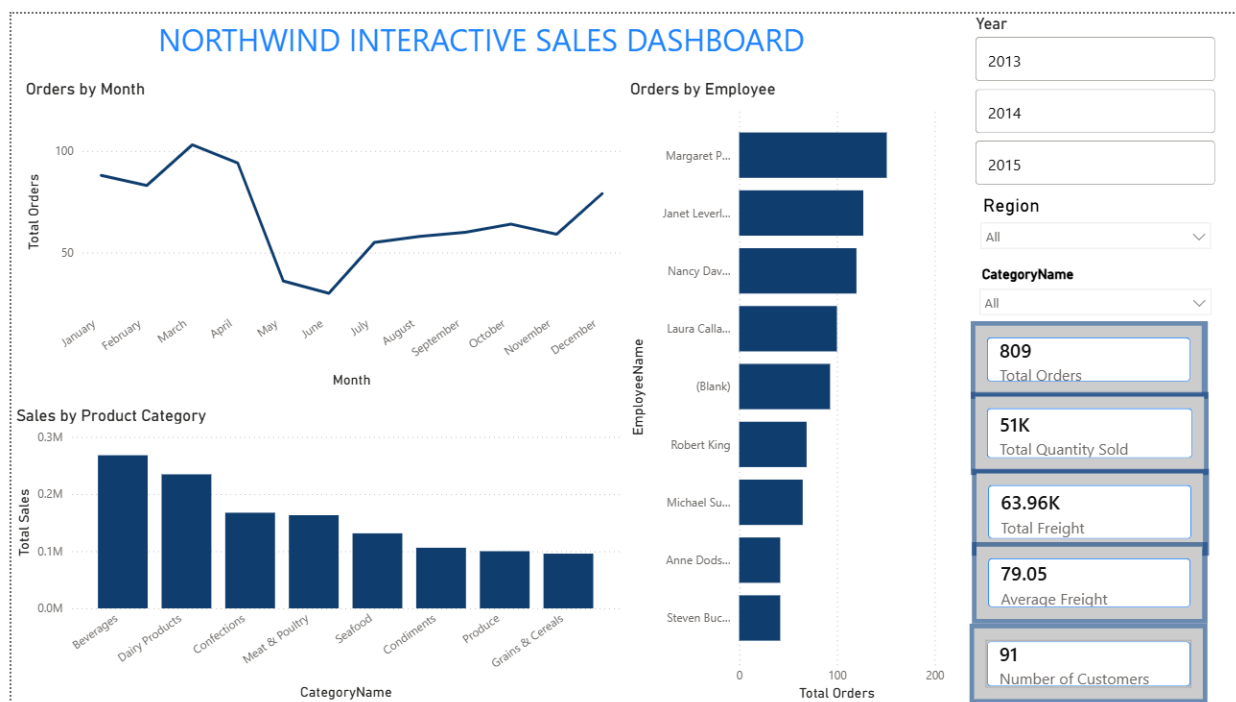
- Orders by Month
- Orders by Employee
- Sales by Product Category
- KPI cards
- Year slicer
- Region slicer

- Category Name slicer

The dashboard allows users to select a particular year, region or category and observe the changes in the visualizations.

Purpose:

- To provide interactive sales analysis.
- To allow users to explore specific segments.
- To compare different periods and categories.
- To make the report more flexible and user-friendly.



→ Category Tooltip

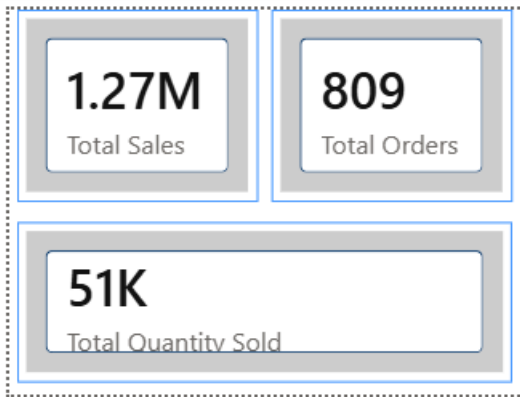
A **Tooltip Page** was created to provide additional information when users interact with a product category.

The tooltip feature allows users to obtain additional category-related information without leaving the main dashboard.

Purpose:

- To provide additional information without overcrowding the dashboard.

- To improve user interaction.
- To make category analysis more informative.



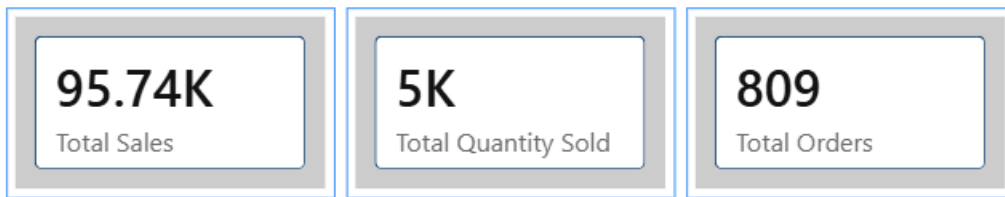
→ Category Details

A separate **Category Details** page is used to provide more detailed information about individual categories.

This allows users to move from the overall category comparison to a more detailed analysis.

Purpose:

- To analyse individual categories in greater detail.
- To provide additional product information.
- To support detailed category-level analysis.



ProductName	QuantityPerUnit	CategoryName	Total Sales	Total Quantity Sold
Filo Mix	16 - 2 kg boxes	Grains & Cereals	3,232.95	500
Gnocchi di nonna Alice	24 - 250 g pkgs.	Grains & Cereals	42,593.06	1263
Gustaf's Knackebröd	24 - 500 g pkgs.	Grains & Cereals	7,122.36	348
Ravioli Angelo	24 - 250 g pkgs.	Grains & Cereals	7,661.55	434
Singaporean Hokkien Fried Mee	32 - 1 kg pkgs.	Grains & Cereals	8,575.00	697
Tunnbröd	12 - 250 g pkgs.	Grains & Cereals	4,601.70	580
Wimmers gute Semmelknödel	20 bags x 4 pieces	Grains & Cereals	21,957.97	740
Total			95,744.59	4562

7. Slicers Used

The **Interactive Sales Dashboard** contains the following slicers:

1. Year

The Year slicer allows users to select:

- 2013
- 2014
- 2015

This allows users to analyse order and sales performance for a particular year.

2. Region

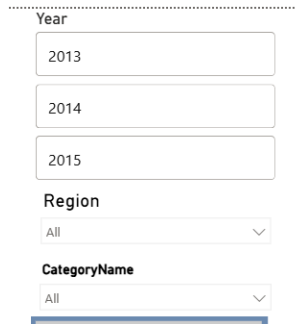
The Region slicer allows users to filter the dashboard according to geographical region.

3. Category Name

The Category Name slicer allows users to select a particular product category.

These slicers interact with the dashboard visualizations and make it possible to perform customized analysis.

For example, selecting a particular year and category allows the user to analyse the monthly orders and sales performance for that specific period and product category.



The image shows a screenshot of a dashboard's filter section. It contains three main filter groups: 'Year', 'Region', and 'CategoryName'. The 'Year' group has three input boxes with the values '2013', '2014', and '2015'. The 'Region' group has a dropdown menu currently set to 'All'. The 'CategoryName' group has a dropdown menu also set to 'All'. The entire filter section is enclosed in a dashed border.

8. Key Findings

The major findings from the Northwind Sales Performance Dashboard are:

- The dashboard displays **809 total orders**.
- Total quantity sold is approximately **51K units**.
- Total freight is approximately **63.96K**.
- Average freight per order is approximately **79.05**.
- The dataset contains **91 customers**.
- Monthly order activity varies throughout the year.
- **March** records the highest monthly order activity in the displayed trend.
- Order activity decreases significantly during **May and June**.
- Orders increase again during the later months of the year.
- **Margaret Peacock** has the highest order count among the employees displayed.
- **Beverages** is the highest-selling product category.
- **Dairy Products** is another major contributor to total sales.
- **Grains & Cereals** has the lowest sales among the displayed categories.
- The **USA** has the largest customer base among the countries shown.
- Product-level analysis shows significant differences in sales and quantity sold between individual products.
- Interactive slicers allow users to analyse the dashboard according to year, region and product category.

9. Importance of the Dashboard

The Northwind Sales Performance Dashboard provides several benefits for business analysis.

Easy Data Understanding

Large amounts of order and product information can be understood through simple charts and KPI cards.

Quick Decision-Making

The KPI cards provide important business information immediately.

Monthly Trend Analysis

The Orders by Month line chart helps identify changes in order activity throughout the year.

Employee Performance Analysis

The Orders by Employee chart helps compare the number of orders handled by employees.

Product Category Analysis

The Sales by Product Category chart helps identify categories contributing more or less to sales.

Customer Analysis

The Customers by Country chart helps identify the geographical distribution of customers.

Product-Level Analysis

The Product & Customer Analysis page provides detailed information about individual products and their sales.

Interactive Analysis

The Year, Region and Category Name slicers allow users to perform customized analysis.

Better Dashboard Experience

Tooltips and detailed category pages provide additional information without making the main dashboard overcrowded.

10. Conclusion

The **Northwind Sales Performance Dashboard** demonstrates how **Microsoft Power BI** can transform raw sales and order data into an interactive and meaningful business intelligence report.

The project helped in understanding important Power BI concepts such as **data preparation, table relationships, Power Query, DAX measures, KPI cards, line charts, bar charts, slicers, tooltips and interactive dashboard design.**

The dashboard provides useful insights into total orders, quantity sold, freight, customer distribution, monthly order trends, employee performance and product category sales.

The use of multiple dashboard pages such as **Sales Overview, Product & Customer Analysis and Interactive Sales Dashboard** makes it possible to analyse the business from different perspectives. The slicers and tooltip features further improve the interactivity of the report.

Overall, the project demonstrates how Power BI can transform Northwind's sales data into clear visual insights and support **effective business analysis and data-driven decision-making.**

11. References

- Microsoft Power BI Desktop
- Northwind Sales Dataset
- Microsoft Power Query

- [Microsoft Power BI Documentation](#)